LEZIONE 4

APPRENDIMENTO SUPERVISIONATO $\longrightarrow$ Machine Learning

$$\left.\begin{array}{l} D_{train} = \{ (\underline{x}, y) : \underline{x} \in \mathbb{R}^N, \ y \in \mathbb{R}^M \} \\ D_{val} \qquad D_{train} \cap D_{val} = \emptyset \\ D_{test} \qquad D_{train} \cap D_{test} = \emptyset \end{array}\right\} \text{DATASET}$$

$\hookrightarrow$ ALGORITMO di ADDESTRAMENTO · Modificare i PARAMETRI $\theta = (W, b)$

$\hookrightarrow$ MODELLO è "PRONTO" e lo si testa su D_{test}

$\hookrightarrow$ INFERENZA

APPRENDIMENTO AUTOMATICO su SLP e MLP a 1 strato nascosto

$\downarrow$

Si basa sul DATASET : OTTIMIZZAZIONE

1. Stabilire un modo per misurare l'ERRORE, o il COSTO
2. Minimizzare il COSTO

ESEMPIO: 1. COSTO · $\| \underbrace{f^*(\underline{x})}_{\text{funzione OBIETTIVO}} - \underbrace{F_\theta(\underline{x})}_{\text{funzione calcolata dal MODELLO}} \|$ $\longleftarrow$ NORMA su $\mathbb{R}^M$: distanza

$$\| \underbrace{y^*}_{\text{dal DATASET}} - \underbrace{y}_{y \text{ calcolato}} \|$$

2. $\min_\theta \| y^* - y \| = \min_\theta \| y^* - F_\theta(\underline{x}) \|$, $\theta = (W, b)$

Def. Si dice FUNZIONE di COSTO o di PERDITA o LOSS una funzione definita su un dataset $D = \{ (\underline{x}, y) : \underline{x} \in \mathbb{R}^N, \ y \in \mathbb{R}^M \}$

ovvero $\mathcal{L} : D \longrightarrow \mathbb{R}$

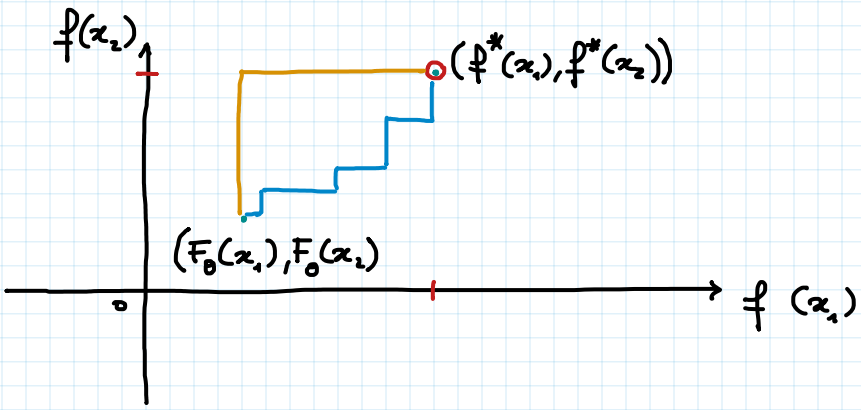
LOSS L_1 $\quad \mathcal{L}_1(D) := \sum_{i=1}^{m} \| f^*(\underline{x}_i) - F_\theta(\underline{x}_i) \| = \sum_{i=1}^{m} \| \underline{y}_i^* - F_\theta(\underline{x}_i) \|$

$|D| = m$

Visualizzazione : $\quad |D| = 2$

$f^* : \mathbb{R} \to \mathbb{R}$

$F_\theta : \mathbb{R} \to \mathbb{R}$



GEOMETRIA del TAXI

LOSS $\mathcal{L}_2$, ERRORE QUADRATICO MEDIO , MEAN-SQUARED ERROR (MSE)

$|D| = m \qquad f^* : \mathbb{R}^N \to \mathbb{R}^M \quad , \quad F_\theta : \mathbb{R}^N \to \mathbb{R}^M$

$$\mathcal{L}_2(D) = \frac{1}{2m} \sum_{i=1}^{m} \| \underbrace{f^*(\underline{x}_i)}_{\underline{y}_i^*} - F_\theta(\underline{x}_i) \|^2$$

$\mathcal{L} : D \to \mathbb{R}$

$\mathcal{L} : \mathbb{R}^P \to \mathbb{R}$

$\theta \mapsto \mathcal{L}(\theta)$

$\mathcal{L}_2$ su SLP $\qquad F_\theta$ è l'SLP

$$\mathcal{L}_2^{SLP}(D) = \frac{1}{2m} \sum_{i=1}^{m} \| \underline{y}_i^* - \sigma(\boxed{W^T} \underline{x}_i + \boxed{\underline{b}}) \|^2 = \mathcal{L}_2^{SLP}(W, \underline{b})$$

$\mathcal{L}_2$ su MLP $\qquad F_\theta$ è l'MLP a 1 strato nascosto

$$\mathcal{L}_2^{MLP}(D) = \frac{1}{2m} \sum_{i=1}^{m} \| \underline{y}_i^* - \sigma(\boxed{W^{(2)}}^T \cdot \sigma(\boxed{W^{(1)}}^T \cdot \underline{x}_i + \boxed{\underline{b}^{(1)}}) + \boxed{\underline{b}^{(2)}}) \|^2$$

$$= \mathcal{L}_2^{MLP}(W^{(1)}, W^{(2)}, \underline{b}^{(1)}, \underline{b}^{(2)})$$

STEP 1: definire il COSTO ✓

STEP 2: MINIMIZZARE il costo

$$\min_{\theta} \mathcal{L}_2^{SLP}(\theta) \quad ; \quad \min_{\theta} \mathcal{L}_2^{MLP}(\theta)$$

$\theta \to W, \underline{b}$; $\theta \to W^{(1)}, W^{(2)}, \underline{b}^{(1)}, \underline{b}^{(2)}$

OSSERVAZIONE: Trovare W e $\underline{b}$ che approx f^*, SE σ è DIFFERENZIABILE non è più MAL-CONDIZIONATO.

$$F_\theta(\underline{x}) \overset{SLP}{=} \sigma(W^T \cdot \underline{x} + \underline{b}) \quad \to \quad F_\theta \text{ è DIFFERENZIABILE}$$

$$dF_\theta^{(j)}(\underline{x}_0) \boxed{=} \sum_{i=1}^{N} \frac{\partial F_\theta}{\partial \cancel{x_i}\,\theta_i} d\cancel{x_i}\,\theta_i = \sum_{i=1}^{N} \underbrace{\frac{\partial \sigma}{\partial w_{ij}}}_{\text{calcolata in } \underline{x}_0 \to \in \mathbb{R}} \boxed{dw_{ij}} + \underbrace{\frac{\partial \sigma}{\partial b_j}}_{\text{calcolata in } \underline{x}_0 \to \in \mathbb{R}} \boxed{db_j}$$

DEF di differenziale

$$dF_{W,\underline{b}}^{(j)}(\underline{x}_0) = \sum_{i=1}^{N} \underline{\lambda}\, dw_{ij} + \underline{\mu}\, db_j$$

STEP 2: MINIMIZZARE il COSTO con l'ALGORITMO di DISCESA del GRADIENTE

$$\begin{cases} \underline{\theta}_0 = a & \text{scelti in modo casuale} \\ \underline{\theta}_k = \underline{\theta}_{k-1} - \eta \underbrace{\nabla_\theta \mathcal{L}}_{\text{GRADIENTE di } \mathcal{L} \text{ fatto rispetto a } \theta \text{ PARAMETRI}}(\underline{\theta}_{k-1}) \end{cases}$$

AGGIORNAMENTO dei PESI, BIAS (PARAMETRI)

η detto COEFFICIENTE di APPRENDIMENTO è un IPERPARAMETRO

Il metodo si conclude dopo un numero N_e di ITERAZIONI / EPOCHE di ADDESTRAMENTO

$$\mathcal{L}^{SLP}(\overset{\theta}{W, \underline{b}}) \geq 0 \quad \text{e vogliamo arrivare a} \quad \mathcal{L}^{SPL}(\overset{\theta}{W, \underline{b}}) = 0$$

$$\mathcal{L}^{SLP}(\underbrace{W, \underline{b}}_{\theta}) \approx 0 \iff F_\theta(\underline{x}) \approx y^* \quad \forall (\underline{x}, y^*) \in D$$

$$\mathcal{L}^{SLP}(\underline{\theta}_k) < \mathcal{L}^{SLP}(\underline{\theta}_{k-1}) \iff \underbrace{\mathcal{L}(\underline{\theta}_k) - \mathcal{L}(\underline{\theta}_{k-1})}_{\Delta\mathcal{L} < 0} < 0 \quad *$$

$$* \quad \boxed{\underline{\theta}_k - \underline{\theta}_{k-1} \approx 0 \implies \Delta\mathcal{L} \approx d\mathcal{L}}$$

$\mathcal{L}$ è DIFFERENZIABILE (in $\mathbb{R}$ equivale a $\mathcal{L}$ DERIVABILE)

$$d\mathcal{L}(\underline{v}) = \nabla_\theta \mathcal{L} \cdot \underline{v} \quad * \quad \text{vogliamo che} \quad \nabla_\theta \mathcal{L} \cdot \underline{v} < 0$$

IDEA:

$$\underline{v} = -\eta \nabla_\theta \mathcal{L} \qquad \eta \in \mathbb{R},\ \eta > 0$$

$$d\mathcal{L}(\underline{v}) = \nabla_\theta \mathcal{L} \cdot (-\eta \cdot \nabla_\theta \mathcal{L}) = -\underbrace{\eta}_{>0} \underbrace{\|\nabla_\theta \mathcal{L}\|^2}_{>0} < 0$$

* Devo fare in modo che l'approssimazione $\Delta\mathcal{L} \approx d\mathcal{L}$ sia BUONA

$\hookrightarrow$ SCELGO η t.c. $\underline{\theta}_k \approx \underline{\theta}_{k-1}$

MA $\|\underline{\theta}_k - \underline{\theta}_{k-1}\| > 0$

η Molto PICCOLO MA non deve ANNULLARE

altrimenti l'algoritmo è fermo

Quindi $\|\underline{v}\| = \varepsilon > 0$

$$\varepsilon = \|\underline{v}\| = \|-\eta \nabla_\theta \mathcal{L}\| \leq \eta \|\nabla_\theta \mathcal{L}\|$$

$$\implies \eta \geq \frac{\varepsilon}{\|\nabla_\theta \mathcal{L}\|}$$

lo possiamo scegliere $\eta = \dfrac{\varepsilon}{\|\nabla_\theta \mathcal{L}\|}$

Def : DERIVATA della FUNZIONE COMPOSTA : CHAIN-RULE

N = 1 $\quad f : \mathbb{R} \to \mathbb{R} \qquad g : \mathbb{R} \to \mathbb{R} \qquad f \circ g : \mathbb{R} \to \mathbb{R} \to \mathbb{R}$

$$x \mapsto g(x) \mapsto f(g(x))$$

$$\frac{df}{dx}(g(x_0)) = \frac{df}{\boxed{dg}}(g(x_0)) \cdot \frac{\boxed{dg}}{dx}(x_0)$$

ESEMPIO : $\quad (2x+3)^2 \underset{\text{derivo}}{\rightsquigarrow} \left[\begin{array}{l} (\cdot)^2 \rightsquigarrow 2\cdot \\ 2\cdot + 3 \rightsquigarrow 2 \end{array}\right] \to 2(2x+3)\cdot 2$

$g \qquad g$

N > 1

1) $\quad f : \underbrace{\mathbb{R}^N} \to \mathbb{R} \quad , \quad g : \mathbb{R} \to \underbrace{\mathbb{R}^N} \qquad f \circ g : \mathbb{R} \to \mathbb{R}^N \to \mathbb{R}$

$(g_1, g_2, \dots, g_N)$

$$x \mapsto g(x) \mapsto f(g(x))$$

2) $\quad f : \underbrace{\mathbb{R}^N} \to \mathbb{R} \quad , \quad g : \mathbb{R}^M \to \underbrace{\mathbb{R}^N} \qquad f \circ g : \mathbb{R}^M \to \mathbb{R}^N \to \mathbb{R}$

$(g_1, g_2, \dots, g_N)$

$$\underline{x} \mapsto g(\underline{x}) \mapsto f(g(\underline{x}))$$

3) $\quad f : \underbrace{\mathbb{R}^N} \to \mathbb{R}^K \quad , \quad g : \mathbb{R}^M \to \underbrace{\mathbb{R}^N} \qquad f \circ g : \mathbb{R}^M \to \mathbb{R}^N \to \mathbb{R}^K$

$(f_1, \dots, f_K) \qquad (g_1, g_2, \dots g_N)$

$$\underline{x} \mapsto g(\underline{x}) \mapsto f(g(\underline{x}))$$

DERIVATA di 1) $\quad f \circ g : \mathbb{R} \to \mathbb{R}^N \to \mathbb{R} \quad : \mathbb{R} \to \mathbb{R}$

$$\frac{df \circ g}{dx}(x_0) = \sum_{i=1}^{N} \frac{\partial f}{\boxed{\partial g_i}}(g_i(x_0)) \cdot \left(\frac{\boxed{dg}}{dx}(x_0)\right)^{\boxed{(i)}} = \nabla_g f(g(x_0)) \cdot \left(\frac{dg}{dx}\right)_{i=1}^{N}$$

prod. scalare

$$\begin{tikzcd}
 & g_1 \arrow[r] & \left(\frac{dg}{dx}\right)_1 \arrow[r, "\times"] & \frac{\partial f}{\partial g_1} \arrow[dr] & & \\
x \arrow[ur] \arrow[r] \arrow[dr] & g_2 \arrow[r] & \left(\frac{dg}{dx}\right)_2 \arrow[r, "\times"] & \frac{\partial f}{\partial g_2} \arrow[r] & + \arrow[r] & y \\
 & \vdots & & \vdots & & \\
 & g_N \arrow[r] & \left(\frac{dg}{dx}\right)_N \arrow[r, "\times"] & \frac{\partial f}{\partial g_N} \arrow[ur] & &
\end{tikzcd}$$

DERIVATA di 2) $f \circ g : \mathbb{R}^M \longrightarrow \mathbb{R}^N \longrightarrow \mathbb{R} \quad \cdot \quad \mathbb{R}^M \longrightarrow \mathbb{R}$

$$\frac{\partial\, f\circ g}{\partial x_k} = \sum_{i=1}^{N} \frac{\partial f}{\partial g_i}(g(\underline{x}_0)) \cdot \frac{\partial g_i}{\partial x_k}(\underline{x}_0)$$

$$\nabla f\circ g = \left(\frac{\partial f\circ g}{\partial x_1}, \frac{\partial f\circ g}{\partial x_2}, \dots, \frac{\partial f\circ g}{\partial x_M} \right)$$

$$g_1 \rightsquigarrow \nabla g_1 = \left(\frac{\partial g_1}{\partial x_1}, \dots, \frac{\partial g_1}{\partial x_M} \right) \xrightarrow{\times} \frac{\partial f}{\partial g_1} \searrow$$

$$g_2 \rightsquigarrow \nabla g_2 = \left(\frac{\partial g_2}{\partial x_1}, \dots, \frac{\partial g_2}{\partial x_M} \right) \xrightarrow{\times} \frac{\partial f}{\partial g_2} \longrightarrow + \rightarrow$$

$$\vdots \qquad\qquad \vdots$$

$$g_N \rightsquigarrow \nabla g_N = \left(\frac{\partial g_N}{\partial x_1}, \dots, \frac{\partial g_N}{\partial x_M} \right) \xrightarrow{\times} \frac{\partial f}{\partial g_N} \nearrow$$

$$x_1 \qquad x_2 \qquad x_M$$

$$\nabla f\circ g = \left(\frac{\partial f\circ g}{\partial x_1}, \frac{\partial f\circ g}{\partial x_2}, \dots, \frac{\partial f\circ g}{\partial x_M} \right)$$

$\longrightarrow \quad y_1, y_2, y_3 \dots, y_M$

MINIMIZZARE $\mathcal{L}_2^{SLP}(W, \underline{b})$ $\quad |D| = m \quad D = \{(\underline{x}^{(i)}, \underline{y}^{(i)})\}$

$$\mathcal{L}_2^{SLP}(W, \underline{b}) = \frac{1}{2m} \sum_{i=1}^{m} \Big\| \overbrace{\sigma(\underbrace{W^T \underline{x}^{(i)} + \underline{b}}_{\text{PRE-ATTIVAZIONE}})}^{②} - \underline{y}^{(i)} \Big\|^2$$

① $s(W, \underline{b}) = W^T \underline{x} + \underline{b}$

ATTIVAZIONE

② $\sigma(s(W, \underline{b}))$

③ $\| \sigma(s(W, \underline{b})) - \underline{y} \|$ NORMA

④ $\| \sigma(s(W, \underline{b})) - \underline{y} \|^2$ $\quad x \longmapsto x^2$

CALCOLARE $\nabla_\theta \mathcal{L} \longrightarrow \nabla_W \mathcal{L}, \ \nabla_{\underline{b}} \mathcal{L}$

$$\nabla_\theta \mathcal{L} = \frac{\partial x^2}{\partial \|\cdot\|} \cdot \underbrace{\frac{\partial \|\cdot\|}{\partial \sigma}}_{③} \cdot \underbrace{\frac{\partial \sigma}{\partial s}}_{②} \cdot \underbrace{\frac{\partial s}{\partial \theta}}_{①}$$

① Rispetto a w_{ij}

$$s : \mathbb{R}^{N \cdot H + H} \longrightarrow \mathbb{R}^H$$

$$(W, \underline{b}) \longmapsto W^T \underline{x} + \underline{b} = \begin{bmatrix} w_{11} \cdot x_1 + w_{21} \cdot x_2 + \dots + w_{N1} x_N + b_1 \\ w_{12} x_1 + w_{22} \cdot x_2 + \dots + w_{N2} x_N + b_2 \\ \vdots \\ w_{1H} x_1 + w_{2H} x_2 + \dots + w_{NH} \cdot x_N + b_H \end{bmatrix}$$

Neurone 1

Neurone 2

Neurone M

Neurone j :

$$w_{1j} x_1 + w_{2j} x_2 + \dots + \boxed{w_{ij}} x_i + \dots + w_{Nj} x_N + b_j$$

$$\frac{\partial s_k}{\partial w_{ij}} = x_i \quad k = j \quad , \quad \frac{\partial s_k}{\partial w_{ij}} = 0 \quad k \neq j$$

INPUT $\quad$ NEURONE

Derivata di ② ATTIVAZIONE $\sigma : \mathbb{R}^M \to \mathbb{R}^M = \begin{bmatrix} \sigma \\ \vdots \\ \sigma \end{bmatrix}$

$\sigma(s_j) : \mathbb{R} \to \mathbb{R}$

$$\frac{d\sigma}{dw_{ij}}(s_k) = 0 \qquad k \neq j$$

dev'essere DERIVABILE

$$\frac{d\sigma}{dw_{ij}}(s_j) = \frac{d\sigma}{ds_j}\,\frac{\partial s_j}{\partial w_{ij}} = \frac{d\sigma}{ds_j}\cdot x_i = \boxed{\sigma'(s_j)}\cdot x_i$$

③ DERIVATA di $\|\cdot\|$:

$$f : \mathbb{R}^M \to \mathbb{R}$$
$$\underline{x} \longmapsto \|\underline{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_M^2}$$

$$\frac{\partial f}{\partial x_i}(\underline{x}) = \frac{x_i}{\|\underline{x}\|}$$

$$\|\sigma(s) - \underline{y}\| : \mathbb{R}^{N\cdot M + M} \longrightarrow \mathbb{R}^M \left(\longrightarrow \mathbb{R}^M\right) \longrightarrow \mathbb{R}$$

Applichiamo la CHAIN-RULE (2^ caso)

se $k \neq j$ $\quad \frac{\partial \sigma(s_k)}{\partial w_{ij}} = 0$

$$\frac{\partial f}{\partial w_{ij}}(\sigma(s) - \underline{y}) = \sum_{k=1}^{M} \frac{\partial f(\sigma(s) - \underline{y})}{\partial \sigma(s_k)} \cdot \frac{\partial \sigma(s_k)}{\partial w_{ij}} =$$

Se $k = j$

$$= \frac{\partial f(\sigma(s) - \underline{y})}{\partial \sigma} \cdot \sigma'(s_j) \cdot x_i =$$

$$= \frac{\sigma(s_j) - y_j}{\|\sigma(s) - \underline{y}\|} \cdot \sigma'(s_j)\; x_i$$